SCALE Induction - Summary

During this induction hackathon, I learnt many new aspects of machine learning and gained hands on experience working with Python modules like pandas, scikit-learn, and matplotlib.

A linear model is a great fit for datasets that have a linear, simple relation between features and target, and can predict outcomes with low error.
During the model training, a contradiction emerged regarding the relationship between `social_media_hours` and `productivity_hours`. When analyzed in isolation, they show a negative correlation, which aligns with human intuition.
However when incorporated into a Multiple Linear Regression model alongside other influencing parameters, the regression coefficient changed its sign and became positive.
The root cause of this was the identical information conveyed by `social_media_hours` and `phone_usage_hours`. The model assigns a strong negative coefficient to total phone usage. To balance this heavy penalty and prevent under-predicting productivity scores, it assigns a positive weight to social media usage as an adjusting counter-weight.

A decision tree model fits better for datasets which are non-linear, and segregated. The depth at which it is trained becomes an important factor in determining its effectiveness at predicting the outcomes.
At high depths, the model starts overfitting to the training data, it keeps splitting the sample space until the leaves are completely pure, effectively memorizing the training noise rather than finding a generalized pattern, and cannot perform well on test data.

For the data provided to us, a Linear Regression model is a better fit because the underlying relationships in the dataset are predominantly linear. While the Decision Tree risks severe overfitting at higher depths by memorizing, a regularized Linear Regression model (or one with redundant features like `social_media_hours` removed) provides much stronger generalization on unseen test data, lower prediction error, and a more robust model.